Inflation Dynamics - Computational Notebook

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Abstract

This computational notebook documents the data construction, model specifications, estimation workflows, and best-model reporting used to study U.S. quarterly inflation dynamics.

This notebook is organized around five pieces.

1. **Data summary.** The data section constructs the effective inflation sample **1969Q2–2022Q4**, with **215 observations** and give statistical summary.
2. **Mean processes.** The mean-process menu includes Constant, ARX(1,1), ARX(2,1), and ARX(2,2) specifications. These define $\hat{\pi}_t$ and therefore the residual sequence passed to volatility models.
3. **GARCH-family volatility models.** The GARCH section estimates GARCH, GJR-GARCH, and EGARCH specifications with normal, standardized Student's t, and Gaussian-mixture residuals.
4. **Regime-switching GARCH.** The regime-switching section studies state-dependent mean dynamics with normal and standardized Student's t residuals.
5. **BEGE-GARCH models.** The BEGE section estimates Constant, Symmetric, InflationDeflation, BadGood, Constant-p Full BEGE, Constant-n Full BEGE, and Full BEGE models.

Notations are defined:

- π_t: realized inflation.
- $\mathcal{F}_{t-1}$: information set available at time $t-1$.
- SPF_{t-1}: professional forecast of inflation π_t based on $\mathcal{F}_{t-1}$.
- $\hat{\pi}_t$: expected inflation of selected mean process, $E[\pi_t \mid \mathcal{F}_{t-1}]$.
- u_t: inflation innovation, defined as $u_t := \pi_t - \hat{\pi}_t$.
- u_t^+: positive residual component, $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$.
- u_t^-: negative residual component, $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$.

The rest of the notebook uses this notation consistently across data summaries, likelihood definitions, estimation scripts, and reported best-model equations.

Computation Notebook

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1. Inflation Summary Statistics

I start with simple statistical summary by plotting inflation and professional forecast over time.

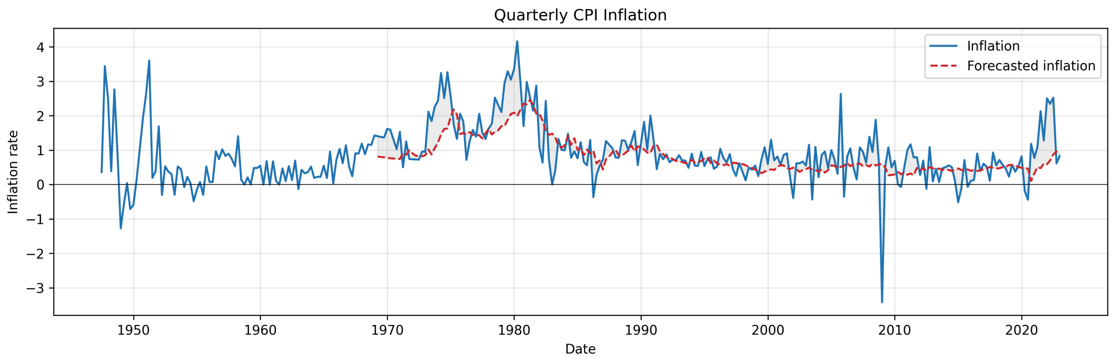


Figure 1: Quarterly inflation π_t and profesional forcast SPF_{t-1}.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Quaterly Inflation Statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use the same effective sample size for all mean models and all volatility models.[1]

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example, σ_0^2, σ_{-1}^2, u_0^2, u_{-1}^2). I use unconditional variance as initialization, for example, in a GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \tag{1}$$

1.b. Effective Sample Summary Statistics

[1]Earlier versions of the estimation code contained a sample-trimming mistake and therefore used different numbers of observations across model settings. For example, some GARCH runs used 208 observations, while some BEGE runs used 210 observations. This coding error has now been corrected: all reported and regenerated estimates use the common **1969Q2–2022Q4** effective sample with **215 observations**. As a result, some current log-likelihood, AIC, and BIC values differ from earlier archived numbers.

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0681	0.3482	−0.7745
P25	0.5328	0.4748	−0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	−0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + u_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + u_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + u_{t+1}$
- **ARX(2,2)** : $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_{t+1}$

where u_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{2}$$

No estimated coefficients. Residuals $u_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: −**222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: −**0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \; \underset{(0.112)}{+0.3005\,\pi_t} \; \underset{(0.167)}{+0.7337\,SPF_t} \tag{3}$$

- Log-likelihood: −**207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: −**1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \; \underset{(0.098)}{+0.2892\,\pi_t} \; \underset{(0.126)}{+0.0834\,\pi_{t-1}} \; \underset{(0.251)}{+0.6385\,SPF_t} \tag{4}$$

- Log-likelihood: −**206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: −**1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \; \underset{(0.106)}{+0.2992\,\pi_t} \; \underset{(0.125)}{+0.0914\,\pi_{t-1}} \; \underset{(0.433)}{+0.4084\,SPF_t} \; \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{5}$$

- Log-likelihood: −**206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: −**1.212**, excess kurtosis: **9.129**

3. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for residuals: normal, Student's t, and a two-component Gaussian mixture.

Let $\{u_t\}_{t=1}^{T}$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. I assume

$$u_t = \sigma_t\, z_t, \qquad z_t \mid \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{6}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or Gaussian mixture) with zero mean and unit variance. Throughout, $z_t = u_t/\sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

3.a. *Volatility Recursion Specifications*

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i\, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k\, \sigma_{t-k}^2, \tag{7}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(u_{t-i}^{+}\right)^2 + \sum_{j=1}^{p} \gamma_j \left(u_{t-j}^{-}\right)^2 + \sum_{k=1}^{q} \beta_k\, \sigma_{t-k}^2, \tag{8}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}\left[\left(u_{t-i}^{+}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \mathbb{E}\left[\left(u_{t-i}^{-}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \frac{1}{2}\sigma_{t-i}^2 \tag{9}$$

, the stationarity condition is

$$\frac{1}{2}\sum_{i=1}^{p} \alpha_i + \frac{1}{2}\sum_{j=1}^{p} \gamma_j + \sum_{k=1}^{q} \beta_k < 1. \tag{10}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(\mid z_{t-i} \mid - \sqrt{\frac{2}{\pi}} \right) + \sum_{j=1}^{p} \gamma_j\, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{11}$$

where $z_t = u_t/\sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{12}$$

3.b. Estimation Initialization And Collection

The GARCH-family estimation is execuated by the `arch` package. For each distribution, mean process, volatility family, and order, the script tries the package default optimizer start and additional randomized feasible starts. The collected result for a model combination is the highest log-likelihood estimate among starts that both converged according to the optimizer and passed the relevant stationarity proxy with a small numerical margin below one (`1e-6`). Fits that fail these checks are excluded from the main result tables and written to a failed-model diagnostic file.

3.c. Log-Likelihood Specifications

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{13}$$

3.c.i. *Normal distribution:*

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0,1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{14}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} \phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\,\sigma_t^2}} \exp\left\{-\frac{u_t^2}{2\,\sigma_t^2}\right\}. \tag{15}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{16}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{17}$$

3.c.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}}\left(1 + \frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \tag{18}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\,\sigma_t^2}}\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{19}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\sigma_t^2\ \right) - \frac{\nu+1}{2}\ln\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{20}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\ \right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right). \tag{21}$$

3.c.iii. *Gaussian mixture distribution:*

The standardized innovation follows a two-component Gaussian mixture. The free parameters are (p_1, μ_1, σ_1^2):

$$f_{\text{mix}}(z) = p_1\,\phi(z;\mu_1,\sigma_1^2) + p_2\,\phi(z;\mu_2,\sigma_2^2), \qquad \phi(z;\mu,s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{22}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\mathrm{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2+\mu_1^2) - p_2\mu_2^2}{p_2}. \tag{23}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\text{mix}}(a) = p_1\,\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\,\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{24}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t} f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{25}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \tag{26}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T}\ln\sigma_t + \sum_{t=1}^{T}\ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right]. \tag{27}$$

4. Model Selection Report

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH[2]

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	ARX(2,1)	(2,1)	387.2314	ARX(1,1)	(2,1)	372.5785	ARX(2,1)	(1,2)	368.6220
Student's t	ARX(2,1)	(1,1)	361.3528	ARX(2,1)	(2,1)	356.9953	ARX(2,1)	(1,2)	353.1752
Gaussian mixture	ARX(1,1)	(2,1)	367.1233	ARX(2,1)	(2,1)	358.4230	ARX(1,1)	(1,2)	353.2873

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	Constant	(2,1)	401.2826	Constant	(2,1)	396.7277	Constant	(2,1)	398.1271
Student's t	Constant	(2,1)	386.6231	Constant	(1,1)	384.2364	Constant	(1,1)	383.5813
Gaussian mixture	ARX(1,1)	(2,1)	400.8297	ARX(1,1)	(1,1)	394.1515	Constant	(1,1)	385.2612

4.a. *Best Models by Criterion and Volatility Family*

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits. The main result tables contain 122 accepted model combinations.

4.a.i. *AIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M073`	Student's t	ARX(2,1)	GARCH(1,1)	−172.6764	361.3528	388.3179

Mean process:

$$\hat{\pi}_{t+1} = 0.0876 + 0.2637\,\pi_t + 0.1716\,\pi_{t-1} + 0.5096\,SPF_t + \mu_{t+1} \tag{28}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0725 + 0.5203\,u_{t-1}^2 + 0.4294\,\sigma_{t-1}^2 \tag{29}$$

Distribution parameters:

[2] These numbers differ from earlier GARCH tables because earlier code used inconsistent effective sample sizes across model settings due to a sample-trimming error. The correction is discussed in the Data Summary effective-sample section. The current estimates use the common **1969Q2–2022Q4** sample with **215 observations**.

Parameter	Estimate
ν	4.3023

Parameter	Coef	Std Err	t-value	p-value
c	0.0876	0.0777	1.1280	0.2593
ρ_1	0.2637	0.0912	2.8914	0.0038
ρ_2	0.1716	0.0975	1.7600	0.0784
ϕ_1	0.5096	0.1487	3.4274	0.0006
ω	0.0725	0.0385	1.8834	0.0596
α_1	0.5203	0.2424	2.1464	0.0318
β_1	0.4294	0.1425	3.0126	0.0026
ν	4.3023	1.4648	2.9372	0.0033

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M080`	Student's t	ARX(2,1)	GJR-GARCH(2,1)	−167.4976	356.9953	394.0723

Mean process:

$$\hat{\pi}_{t+1} = 0.0901 + 0.2045\,\pi_t + 0.1144\,\pi_{t-1} + 0.6522\,SPF_t + \mu_{t+1} \tag{30}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.1096 + 0.4005\left(u_{t-1}^{+}\right)^2 + 0.2848\left(u_{t-1}^{-}\right)^2 + 0.6415\left(u_{t-2}^{+}\right)^2 - 0.0000\left(u_{t-2}^{-}\right)^2 + 0.1559\,\sigma_{t-1}^2 \tag{31}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	5.3421

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0901	0.0838	1.0747	0.2825
ρ_1	0.2045	0.0960	2.1291	0.0332
ρ_2	0.1144	0.1043	1.0969	0.2727
ϕ_1	0.6522	0.1740	3.7481	0.0002
ω	0.1096	0.0397	2.7594	0.0058
α_1	0.4005	0.2194	1.8250	0.0680

Parameter	Coef	Std Err	t-value	p-value
$\gamma_1 - \alpha_1$	−0.1157	0.2773	−0.4172	0.6765
α_2	0.6415	0.3571	1.7963	0.0725
$\gamma_2 - \alpha_2$	−0.6415	0.3194	−2.0082	0.0446
β_1	0.1559	0.1499	1.0399	0.2984
ν	5.3421	1.9529	2.7354	0.0062

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M078`	Student's t	ARX(2,1)	EGARCH(1,2)	−166.5876	353.1752	386.8816

Mean process:

$$\hat{\pi}_{t+1} = 0.0496 + 0.0937\,\pi_t + 0.0735\,\pi_{t-1} + 0.9380\,SPF_t + \mu_{t+1} \tag{32}$$

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.0946 - 0.3695\left(|\ z_{t-1}\ | - \sqrt{2/\pi}\right) + 0.2773\,z_{t-1} + 0.0679\ln\sigma_{t-1}^2 + 0.8601\ln\sigma_{t-2}^2 \tag{33}$$

Distribution parameters:

Parameter	Estimate
ν	5.2826

Parameter	Coef	Std Err	t-value	p-value
c	0.0496	0.0466	1.0655	0.2867
ρ_1	0.0937	0.0833	1.1251	0.2606
ρ_2	0.0735	0.0440	1.6718	0.0946
ϕ_1	0.9380	0.0000	36243962.4808	0.0000
ω	−0.0946	0.0001	−1626.9937	0.0000
α_1	−0.3695	0.0003	−1381.2872	0.0000
γ_1	0.2773	0.0112	24.7779	0.0000
β_1	0.0679	0.0084	8.0880	0.0000
β_2	0.8601	0.0000	1485807.1736	0.0000
ν	5.2826	1.7878	2.9548	0.0031

4.a.ii. *BIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M055`	Student's t	Constant	GARCH(2,1)	−179.8850	369.7699	386.6231

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{34}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.1192 + 0.4570\, u_{t-1}^2 + 0.5209\, u_{t-2}^2 + 0.0000\, \sigma_{t-1}^2 \tag{35}$$

Distribution parameters:

Parameter	Estimate
ν	4.9417

Parameter	Coef	Std Err	t-value	p-value
ω	0.1192	0.0417	2.8561	0.0043
α_1	0.4570	0.1774	2.5755	0.0100
α_2	0.5209	0.2133	2.4419	0.0146
β_1	0.0000	0.1189	0.0000	1.0000
ν	4.9417	1.7105	2.8890	0.0039

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M050`	Student's t	Constant	GJR-GARCH(1,1)	−178.6916	367.3832	384.2364

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{36}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.0636 + 0.7969\, \left(u_{t-1}^{+}\right)^2 + 0.1469\, \left(u_{t-1}^{-}\right)^2 + 0.4353\, \sigma_{t-1}^2 \tag{37}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	4.7916

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value

Parameter	Coef	Std Err	t-value	p-value
ω	0.0636	0.0255	2.4925	0.0127
α_1	0.7969	0.2566	3.1058	0.0019
$\gamma_1 - \alpha_1$	−0.6499	0.2320	−2.8009	0.0051
β_1	0.4353	0.1168	3.7262	0.0002
ν	4.7916	1.5515	3.0883	0.0020

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M051`	Student's t	Constant	EGARCH(1,1)	−178.3641	366.7281	383.5813

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{38}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.2496 + 0.5580 \left(\mid z_{t-1} \mid - \sqrt{2/\pi} \right) + 0.2647\, z_{t-1} + 0.7897 \ln \sigma_{t-1}^2 \tag{39}$$

Distribution parameters:

Parameter	Estimate
ν	5.0826

Parameter	Coef	Std Err	t-value	p-value
ω	−0.2496	0.1063	−2.3491	0.0188
α_1	0.5580	0.1395	4.0011	0.0001
γ_1	0.2647	0.0782	3.3870	0.0007
β_1	0.7897	0.0610	12.9553	0.0000
ν	5.0826	1.7272	2.9427	0.0033

4.b. *GARCH(1,1) and GJR-GARCH(1,1) Comparisons*

These estimates are reported by mean process and innovation distribution because the GARCH(1,1) and GJR-GARCH(1,1) recursions are useful benchmarks for the BEGE specifications.

4.b.i. *Constant:*

Normal:

GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M001`	Normal	Constant	GARCH(1,1)	−198.2015	402.4029	412.5148

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{40}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.0694 + 0.4799\, u_{t-1}^2 + 0.4725\, \sigma_{t-1}^2 \tag{41}$$

Parameter	Coef	Std Err	t-value	p-value
ω	0.0694	0.0401	1.7321	0.0833
α_1	0.4799	0.1653	2.9029	0.0037
β_1	0.4725	0.0979	4.8248	0.0000

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M002`	Normal	Constant	GJR-GARCH(1,1)	−189.8927	387.7854	401.2680

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{42}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.0573 + 0.5990\, (u_{t-1}^+)^2 + 0.0194\, (u_{t-1}^-)^2 + 0.5578\, \sigma_{t-1}^2 \tag{43}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
ω	0.0573	0.0223	2.5676	0.0102
α_1	0.5990	0.3145	1.9049	0.0568
$\gamma_1 - \alpha_1$	−0.5795	0.2577	−2.2489	0.0245
β_1	0.5578	0.1685	3.3096	0.0009

Student's t:

GARCH(1,1):

No successful model.

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M050`	Student's t	Constant	GJR-GARCH(1,1)	−178.6916	367.3832	384.2364

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{44}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.0636 + 0.7969\left(u_{t-1}^{+}\right)^2 + 0.1469\left(u_{t-1}^{-}\right)^2 + 0.4353\,\sigma_{t-1}^2 \tag{45}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	4.7916

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
ω	0.0636	0.0255	2.4925	0.0127
α_1	0.7969	0.2566	3.1058	0.0019
$\gamma_1 - \alpha_1$	−0.6499	0.2320	−2.8009	0.0051
β_1	0.4353	0.1168	3.7262	0.0002
ν	4.7916	1.5515	3.0883	0.0020

4.b.ii. *ARX(1,1):*

Normal:

GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M013`	Normal	ARX(1,1)	GARCH(1,1)	−191.0320	394.0640	414.2879

Mean process:

$$\hat{\pi}_{t+1} = 0.1137 + 0.4783\,\pi_t + 0.4677\,SPF_t + \mu_{t+1} \tag{46}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0978 + 0.7142\,u_{t-1}^2 + 0.2710\,\sigma_{t-1}^2 \tag{47}$$

Parameter	Coef	Std Err	t-value	p-value

c	0.1137	0.0735	1.5469	0.1219
Parameter	Coef	Std Err	t-value	p-value
ρ_1	0.4783	0.2460	1.9445	0.0518
ϕ_1	0.4677	0.2487	1.8806	0.0600
ω	0.0978	0.0460	2.1266	0.0335
α_1	0.7142	0.5291	1.3499	0.1771
β_1	0.2710	0.2284	1.1869	0.2353

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M014`	Normal	ARX(1,1)	GJR-GARCH(1,1)	−183.8229	381.6458	405.2403

Mean process:

$$\hat{\pi}_{t+1} = 0.0840 + 0.2067\,\pi_t + 0.7961\,SPF_t + \mu_{t+1} \tag{48}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0580 + 0.6052\,\left(u_{t-1}^{+}\right)^2 + 0.0035\,\left(u_{t-1}^{-}\right)^2 + 0.6048\,\sigma_{t-1}^2 \tag{49}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0840	0.1082	0.7764	0.4375
ρ_1	0.2067	0.0925	2.2336	0.0255
ϕ_1	0.7961	0.1511	5.2670	0.0000
ω	0.0580	0.0519	1.1174	0.2638
α_1	0.6052	0.5587	1.0832	0.2787
$\gamma_1 - \alpha_1$	−0.6017	0.4985	−1.2070	0.2274
β_1	0.6048	0.3484	1.7358	0.0826

Student's t:

GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M061`	Student's t	ARX(1,1)	GARCH(1,1)	−174.9174	363.8348	387.4293

Mean process:

$$\hat{\pi}_{t+1} = 0.0827 + 0.2800\,\pi_t + 0.6858\,SPF_t + \mu_{t+1} \tag{50}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0910 + 0.5462\, u_{t-1}^2 + 0.3726\, \sigma_{t-1}^2 \tag{51}$$

Distribution parameters:

Parameter	Estimate
ν	4.1326

Parameter	Coef	Std Err	t-value	p-value
c	0.0827	0.0743	1.1132	0.2656
ρ_1	0.2800	0.0999	2.8028	0.0051
ϕ_1	0.6858	0.1126	6.0885	0.0000
ω	0.0910	0.0426	2.1349	0.0328
α_1	0.5462	0.2386	2.2895	0.0221
β_1	0.3726	0.1329	2.8042	0.0050
ν	4.1326	1.2722	3.2484	0.0012

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M062`	Student's t	ARX(1,1)	GJR-GARCH(1,1)	−171.2540	358.5080	385.4731

Mean process:

$$\hat{\pi}_{t+1} = 0.1091 + 0.2377\, \pi_t + 0.7159\, SPF_t + \mu_{t+1} \tag{52}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0698 + 0.7129\, \left(u_{t-1}^{+}\right)^2 + 0.0914\, \left(u_{t-1}^{-}\right)^2 + 0.4892\, \sigma_{t-1}^2 \tag{53}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	4.6954

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.1091	0.0771	1.4159	0.1568
ρ_1	0.2377	0.0931	2.5532	0.0107
ϕ_1	0.7159	0.1146	6.2455	0.0000

ω	0.0698	0.0258	2.7016	0.0069
Parameter	Coef	Std Err	t-value	p-value
α_1	0.7129	0.3163	2.2537	0.0242
$\gamma_1 - \alpha_1$	−0.6215	0.2629	−2.3637	0.0181
β_1	0.4892	0.1522	3.2139	0.0013
ν	4.6954	1.5763	2.9787	0.0029

Gaussian mixture:

GARCH(1,1):

No successful model.

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M110`	Gaussian mixture	ARX(1,1)	GJR-GARCH(1,1)	−170.2226	360.4452	394.1515

Mean process:

$$\hat{\pi}_{t+1} = 0.1002 + 0.2332\,\pi_t + 0.7215\,SPF_t + \mu_{t+1} \tag{54}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0775 + 0.7766\,(u_{t-1}^+)^2 + 0.1301\,(u_{t-1}^-)^2 + 0.4604\,\sigma_{t-1}^2 \tag{55}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
p_1	0.9528
μ_1	0.0316
σ_1^2	0.6599
p_2	0.0472
μ_2	−0.6370
σ_2^2	7.4387

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.1002	0.0729	1.3749	0.1692

ρ_1	0.2332	0.1055	2.2112	0.0270
Parameter	Coef	Std Err	t-value	p-value
ϕ_1	0.7215	0.1141	6.3256	0.0000
ω	0.0775	0.0486	1.5942	0.1109
α_1	0.7766	0.3535	2.1972	0.0280
$\gamma_1 - \alpha_1$	−0.6465	0.2751	−2.3496	0.0188
β_1	0.4604	0.1843	2.4986	0.0125
p_1	0.9528	0.0608	15.6677	0.0000
μ_1	0.0316	0.0586	0.5386	0.5902
σ_1^2	0.6599	0.2142	3.0803	0.0021

4.b.iii. *ARX(2,1):*

Normal:

GARCH(1,1):

No successful model.

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M026`	Normal	ARX(2,1)	GJR-GARCH(1,1)	−183.7897	383.5794	410.5445

Mean process:

$$\hat{\pi}_{t+1} = 0.0828 + 0.2078\,\pi_t + 0.0096\,\pi_{t-1} + 0.7866\,SPF_t + \mu_{t+1} \tag{56}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0571 + 0.6023\left(u_{t-1}^+\right)^2 + 0.0048\left(u_{t-1}^-\right)^2 + 0.6090\,\sigma_{t-1}^2 \tag{57}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0828	0.1091	0.7592	0.4478
ρ_1	0.2078	0.0927	2.2413	0.0250
ρ_2	0.0096	0.1014	0.0946	0.9246
ϕ_1	0.7866	0.2178	3.6120	0.0003
ω	0.0571	0.0547	1.0441	0.2964
α_1	0.6023	0.5989	1.0057	0.3146
$\gamma_1 - \alpha_1$	−0.5975	0.5362	−1.1144	0.2651

β_1	0.6090	0.3734	1.6307	0.1029

Student's t:

GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M073`	Student's t	ARX(2,1)	GARCH(1,1)	−172.6764	361.3528	388.3179

Mean process:

$$\hat{\pi}_{t+1} = 0.0876 + 0.2637\,\pi_t + 0.1716\,\pi_{t-1} + 0.5096\,SPF_t + \mu_{t+1} \tag{58}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0725 + 0.5203\,u_{t-1}^2 + 0.4294\,\sigma_{t-1}^2 \tag{59}$$

Distribution parameters:

Parameter	Estimate
ν	4.3023

Parameter	Coef	Std Err	t-value	p-value
c	0.0876	0.0777	1.1280	0.2593
ρ_1	0.2637	0.0912	2.8914	0.0038
ρ_2	0.1716	0.0975	1.7600	0.0784
ϕ_1	0.5096	0.1487	3.4274	0.0006
ω	0.0725	0.0385	1.8834	0.0596
α_1	0.5203	0.2424	2.1464	0.0318
β_1	0.4294	0.1425	3.0126	0.0026
ν	4.3023	1.4648	2.9372	0.0033

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M074`	Student's t	ARX(2,1)	GJR-GARCH(1,1)	−170.0718	358.1436	388.4794

Mean process:

$$\hat{\pi}_{t+1} = 0.1086 + 0.2335\,\pi_t + 0.1251\,\pi_{t-1} + 0.5871\,SPF_t + \mu_{t+1} \tag{60}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0611 + 0.6588\,\left(u_{t-1}^+\right)^2 + 0.1271\,\left(u_{t-1}^-\right)^2 + 0.5231\,\sigma_{t-1}^2 \tag{61}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
Parameter	Estimate
ν	4.5950

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.1086	0.0791	1.3737	0.1695
ρ_1	0.2335	0.0872	2.6767	0.0074
ρ_2	0.1251	0.0962	1.3002	0.1935
ϕ_1	0.5871	0.1515	3.8739	0.0001
ω	0.0611	0.0282	2.1700	0.0300
α_1	0.6588	0.3260	2.0205	0.0433
$\gamma_1 - \alpha_1$	−0.5316	0.2567	−2.0711	0.0384
β_1	0.5231	0.1823	2.8688	0.0041
ν	4.5950	1.5577	2.9498	0.0032

Gaussian mixture:

GARCH(1,1):

No successful model.

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M122`	Gaussian mixture	ARX(2,1)	GJR-GARCH(1,1)	−168.6131	359.2261	396.3031

Mean process:

$$\hat{\pi}_{t+1} = 0.1039 + 0.2327\,\pi_t + 0.1491\,\pi_{t-1} + 0.5478\,SPF_t + \mu_{t+1} \tag{62}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0720 + 0.8102\left(u_{t-1}^+\right)^2 + 0.2212\left(u_{t-1}^-\right)^2 + 0.4566\,\sigma_{t-1}^2 \tag{63}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
p_1	0.9629

μ_1	0.0278
Parameter	Estimate
σ_1^2	0.6478
p_2	0.0371
μ_2	−0.7218
σ_2^2	9.5904

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.1039	0.0685	1.5176	0.1291
ρ_1	0.2327	0.0994	2.3402	0.0193
ρ_2	0.1491	0.0954	1.5630	0.1180
ϕ_1	0.5478	0.1606	3.4108	0.0006
ω	0.0720	0.0500	1.4414	0.1495
α_1	0.8102	0.3943	2.0550	0.0399
$\gamma_1 - \alpha_1$	−0.5889	0.2797	−2.1056	0.0352
β_1	0.4566	0.1677	2.7223	0.0065
p_1	0.9629	0.0454	21.2086	0.0000
μ_1	0.0278	0.0575	0.4838	0.6285
σ_1^2	0.6478	0.2282	2.8394	0.0045

4.b.iv. *ARX(2,2):*

Normal:

GARCH(1,1):

No successful model.

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M038`	Normal	ARX(2,2)	GJR-GARCH(1,1)	−183.7821	385.5642	415.9000

Mean process:

$$\hat{\pi}_{t+1} = 0.0822 + 0.2087\,\pi_t + 0.0102\,\pi_{t-1} + 0.7570\,SPF_t + 0.0279\,SPF_{t-1} + \mu_{t+1} \tag{64}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0566 + 0.5988\left(u_{t-1}^+\right)^2 + 0.0044\left(u_{t-1}^-\right)^2 + 0.6116\,\sigma_{t-1}^2 \tag{65}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0822	0.1120	0.7342	0.4628
ρ_1	0.2087	0.0935	2.2310	0.0257
ρ_2	0.0102	0.1021	0.1000	0.9204
ϕ_1	0.7570	0.4131	1.8325	0.0669
ϕ_2	0.0279	0.3190	0.0874	0.9304
ω	0.0566	0.0580	0.9757	0.3292
α_1	0.5988	0.6262	0.9563	0.3389
$\gamma_1 - \alpha_1$	−0.5945	0.5631	−1.0557	0.2911
β_1	0.6116	0.3941	1.5518	0.1207

Student's t:

GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M085`	Student's t	ARX(2,2)	GARCH(1,1)	−172.5010	363.0019	393.3377

Mean process:

$$\hat{\pi}_{t+1} = 0.0841 + 0.2681\,\pi_t + 0.1768\,\pi_{t-1} + 0.3222\,SPF_t + 0.1807\,SPF_{t-1} + \mu_{t+1} \tag{66}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0730 + 0.5164\,u_{t-1}^2 + 0.4303\,\sigma_{t-1}^2 \tag{67}$$

Distribution parameters:

Parameter	Estimate
ν	4.2836

Parameter	Coef	Std Err	t-value	p-value
c	0.0841	0.0798	1.0530	0.2924
ρ_1	0.2681	0.0925	2.8982	0.0038
ρ_2	0.1768	0.0989	1.7884	0.0737
ϕ_1	0.3222	0.3874	0.8316	0.4057
ϕ_2	0.1807	0.3505	0.5156	0.6061
ω	0.0730	0.0408	1.7903	0.0734
α_1	0.5164	0.2519	2.0501	0.0404
β_1	0.4303	0.1615	2.6639	0.0077

ν	4.2836	1.4494	2.9554	0.0031

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M086`	Student's t	ARX(2,2)	GJR-GARCH(1,1)	−169.9900	359.9800	393.6864

Mean process:

$$\hat{\pi}_{t+1} = 0.1061 + 0.2373\,\pi_t + 0.1287\,\pi_{t-1} + 0.4591\,SPF_t + 0.1216\,SPF_{t-1} + \mu_{t+1} \tag{68}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0622 + 0.6620\,(u_{t-1}^+)^2 + 0.1336\,(u_{t-1}^-)^2 + 0.5170\,\sigma_{t-1}^2 \tag{69}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	4.5814

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.1061	0.0810	1.3095	0.1904
ρ_1	0.2373	0.0892	2.6599	0.0078
ρ_2	0.1287	0.0980	1.3140	0.1888
ϕ_1	0.4591	0.4103	1.1191	0.2631
ϕ_2	0.1216	0.3701	0.3284	0.7426
ω	0.0622	0.0301	2.0638	0.0390
α_1	0.6620	0.3330	1.9881	0.0468
$\gamma_1 - \alpha_1$	−0.5284	0.2661	−1.9860	0.0470
β_1	0.5170	0.1907	2.7106	0.0067
ν	4.5814	1.5473	2.9609	0.0031

Gaussian mixture:

GARCH(1,1):

No successful model.

GJR-GARCH(1,1):

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M134`	Gaussian mixture	ARX(2,2)	GJR-GARCH(1,1)	−168.4801	360.9602	401.4079

Mean process:

$$\hat{\pi}_{t+1} = 0.1043 + 0.2348\,\pi_t + 0.1514\,\pi_{t-1} + 0.3929\,SPF_t + 0.1444\,SPF_{t-1} + \mu_{t+1} \tag{70}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0729 + 0.8250\,(u_{t-1}^{+})^2 + 0.2282\,(u_{t-1}^{-})^2 + 0.4512\,\sigma_{t-1}^2 \tag{71}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
p_1	0.9611
μ_1	0.0299
σ_1^2	0.6371
p_2	0.0389
μ_2	−0.7376
σ_2^2	9.4010

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.1043	0.0681	1.5304	0.1259
ρ_1	0.2348	0.0990	2.3724	0.0177
ρ_2	0.1514	0.0948	1.5959	0.1105
ϕ_1	0.3929	0.3600	1.0914	0.2751
ϕ_2	0.1444	0.3070	0.4703	0.6381
ω	0.0729	0.0494	1.4773	0.1396
α_1	0.8250	0.3999	2.0631	0.0391
$\gamma_1 - \alpha_1$	−0.5968	0.2880	−2.0721	0.0383
β_1	0.4512	0.1784	2.5297	0.0114
p_1	0.9611	0.0445	21.5747	0.0000
μ_1	0.0299	0.0597	0.5001	0.6170

σ_1^2	0.6371	0.2281	2.7931	0.0052

4.c. Excluded Model Combinations

The combinations below did not produce an optimizer-converged fit satisfying the stationarity proxy with the `1e-6` margin after the restart search. They are omitted from the model-selection tables and retained in `garch_family_failed_models.csv` for audit.

Model ID	Distribution	Mean process	Volatility process	Attempts	Best attempted LogLik
`M025`	Normal	ARX(2,1)	GARCH(1,1)	50	−188.8527
`M028`	Normal	ARX(2,1)	GARCH(1,2)	50	−188.8527
`M037`	Normal	ARX(2,2)	GARCH(1,1)	50	−188.8355
`M040`	Normal	ARX(2,2)	GARCH(1,2)	50	−188.8355
`M049`	Student's t	Constant	GARCH(1,1)	50	−181.9942
`M052`	Student's t	Constant	GARCH(1,2)	50	−181.9942
`M097`	Gaussian mixture	Constant	GARCH(1,1)	50	−179.4873
`M098`	Gaussian mixture	Constant	GJR-GARCH(1,1)	50	−175.2123
`M100`	Gaussian mixture	Constant	GARCH(1,2)	50	−179.4873
`M101`	Gaussian mixture	Constant	GJR-GARCH(1,2)	50	−175.2123
`M103`	Gaussian mixture	Constant	GARCH(2,1)	50	−177.7161
`M104`	Gaussian mixture	Constant	GJR-GARCH(2,1)	50	−171.8210
`M106`	Gaussian mixture	Constant	GARCH(2,2)	50	−177.7161
`M107`	Gaussian mixture	Constant	GJR-GARCH(2,2)	50	−171.8210
`M109`	Gaussian mixture	ARX(1,1)	GARCH(1,1)	50	−174.5852
`M112`	Gaussian mixture	ARX(1,1)	GARCH(1,2)	50	−174.5852
`M121`	Gaussian mixture	ARX(2,1)	GARCH(1,1)	50	−171.3839
`M124`	Gaussian mixture	ARX(2,1)	GARCH(1,2)	50	−171.3839
`M127`	Gaussian mixture	ARX(2,1)	GARCH(2,1)	50	−170.7809
`M130`	Gaussian mixture	ARX(2,1)	GARCH(2,2)	50	−170.6186
`M133`	Gaussian mixture	ARX(2,2)	GARCH(1,1)	50	−171.1994
`M139`	Gaussian mixture	ARX(2,2)	GARCH(2,1)	50	−170.7011

4.d. EGARCH and Initialization Notes

- Effective sample is fixed at 215 observations for all models (`hold_back=0` with explicit lag regressors in the mean equation).
- In `arch`, EGARCH uses the package's centered term `|z|-sqrt(2/pi)` in the recursion for all distributions.
- Under non-Gaussian errors (e.g., Student's t or Gaussian mixture), this is an intercept reparameterization; fitted dynamics and likelihood are still valid.
- Conditional-variance recursion starts use the default initialization implemented by the `arch` package; this report no longer iterates the `backcast` to the parameter-implied unconditional variance.

- Each model combination is estimated from the package default start plus randomized feasible starts, and only optimizer-converged fits that pass the stationarity proxy with a `1e-6` margin below one are collected in the main result tables.
- Stability is monitored using a persistence proxy (`< 1 - 1e-6`): GARCH uses `sum(alpha)+sum(beta)`, GJR uses `sum(alpha)+0.5*sum(gamma)+sum(beta)`, EGARCH uses `sum(beta)`.

5. Regime-Switching Inflation Models

Let the latent state be

$$s_t \in \{1, ..., K\}, \qquad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \tag{72}$$

I estimate several specifications that differ in **which blocks of parameters are allowed to switch with s_t**. Taking the ARX(2,2) model as an example,

$$\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 \,\mathrm{SPF}_t + \phi_2 \,\mathrm{SPF}_{t-1} + \mu_{t+1}, \tag{73}$$

the parameters are grouped into three blocks, and each block can independently be made regime-dependent:

Block	State-dependent parameters
AR component	$c_{s_t},\ \rho_{1,s_t},\ \rho_{2,s_t}$
SPF component	$\phi_{1,s_t},\ \phi_{2,s_t}$
Shock distribution	$\sigma_{s_t},\ \nu_{s_t}$

where ν_{s_t} denotes the degrees of freedom when assuming standardized Student-t distribution for residuals. A given specification is defined by selecting any subset of these three blocks to be switching; parameters in blocks not selected are held constant across regimes.

Mean	Dist	K	Sw.AR	Sw.SPF	Sw.Var	Sw.nu	LogLik	AIC	BIC
Constant	Normal	2	N	N	Y	N	−186.63	381.26	394.743
Constant	Normal	3	N	N	Y	N	−180.655	379.311	409.646
Constant	Student t	2	N	N	Y	Y	−185.7	383.4	403.624
Constant	Student t	3	N	N	Y	Y	−180.389	384.778	425.226
ARX(1,1)	Normal	2	N	N	Y	N	−177.189	368.378	391.972
ARX(1,1)	Normal	2	Y	N	Y	N	−175.118	368.235	398.571
ARX(1,1)	Normal	2	Y	Y	Y	N	−175.092	370.185	403.891
ARX(1,1)	Normal	3	N	N	Y	N	−173.724	371.448	411.896
ARX(1,1)	Normal	3	Y	N	Y	N	−162.374	356.747	410.677
ARX(1,1)	Student t	2	N	N	Y	Y	−175.494	368.988	399.324
ARX(1,1)	Student t	2	Y	N	Y	Y	−162.972	347.945	385.022
ARX(1,1)	Student t	2	Y	Y	Y	Y	−162.961	349.921	390.369
ARX(1,1)	Student t	3	N	N	Y	Y	−173.091	376.182	426.742
ARX(1,1)	Student t	3	Y	N	Y	Y	−155.151	348.301	412.343
ARX(2,1)	Normal	2	N	N	Y	N	−176.216	368.431	395.396
ARX(2,1)	Normal	2	Y	N	Y	N	−174.942	371.884	408.961
ARX(2,1)	Normal	2	Y	Y	Y	N	−174.377	372.754	413.201
ARX(2,1)	Normal	3	N	N	Y	N	−171.151	368.301	412.119
ARX(2,1)	Normal	3	Y	N	Y	N	−156.72	351.44	415.482
ARX(2,1)	Student t	2	N	N	Y	Y	−173.425	366.849	400.556
ARX(2,1)	Student t	2	Y	N	Y	Y	−160.571	347.142	390.96
ARX(2,1)	Student t	2	Y	Y	Y	Y	−159.534	347.068	394.257
ARX(2,1)	Student t	3	N	N	Y	Y	−171.069	374.139	428.069
ARX(2,1)	Student t	3	Y	N	Y	Y	−150.126	344.253	418.407
ARX(2,2)	Normal	2	N	N	Y	N	−175.961	369.922	400.258
ARX(2,2)	Normal	2	Y	N	Y	N	−174.534	373.067	413.515
ARX(2,2)	Normal	2	Y	Y	Y	N	−174.155	376.31	423.499
ARX(2,2)	Normal	3	N	N	Y	N	−171.029	370.057	417.246
ARX(2,2)	Normal	3	Y	N	Y	N	−156.663	353.325	420.738
ARX(2,2)	Student t	2	N	N	Y	Y	−173.233	368.466	405.543
ARX(2,2)	Student t	2	Y	N	Y	Y	−160.567	349.135	396.324
ARX(2,2)	Student t	2	Y	Y	Y	Y	−157.588	347.176	401.107
ARX(2,2)	Student t	3	N	N	Y	Y	−170.754	375.507	432.808
ARX(2,2)	Student t	3	Y	N	Y	Y	−149.466	344.932	422.457

6. Best Regime-Switching Model

Selected by AIC among converged models.

6.a. Model Summary

- Model label: `ARX(2,1)_student_t_r3_arY_spfN`
- Mean process: `ARX(2,1)`
- Mean equation target: `Inflation`
- Intercept included: `Y`
- Intercept switches by regime: `Y`
- Error distribution: `Student t`
- #Regime: `3`
- Switching AR: `Y`
- Switching SPF: `N`
- Switching distribution variance: `Y`
- Switching nu: `Y`
- Sample: `1969Q2 to 2022Q4`
- Nobs: `215`
- LogLik: `-150.126`
- AIC: `344.253`
- BIC: `418.407`

6.b. Mean Model Specification

```
Inflation = c_(s_t) + beta_{Inflation_lag_1}(s_t) * Inflation_lag_1 +
beta_{Inflation_lag_2}(s_t) * Inflation_lag_2 + beta_{SPF} * SPF + u_t
```

Regime-specific mean coefficients:

regime	Intercept	Inflation_lag_1	Inflation_lag_2	SPF
0	0.872769	−0.54292	−0.238057	0.574951
1	0.381796	0.0492982	−0.0989449	0.574951
2	−0.0670625	0.26872	0.448281	0.574951

6.c. Mean Process Parameters

parameter	estimate
const[0]	0.872769
const[1]	0.381796
const[2]	−0.0670625
Inflation_lag_1[0]	−0.54292
Inflation_lag_1[1]	0.0492982
Inflation_lag_1[2]	0.26872
Inflation_lag_2[0]	−0.238057
Inflation_lag_2[1]	−0.0989449
Inflation_lag_2[2]	0.448281

SPF	0.574951

6.d. Distribution Parameters

parameter	estimate
sigma2[0]	0.0118058
sigma2[1]	0.612973
sigma2[2]	0.208789
nu[0]	183.071
nu[1]	2.39744
nu[2]	104.326

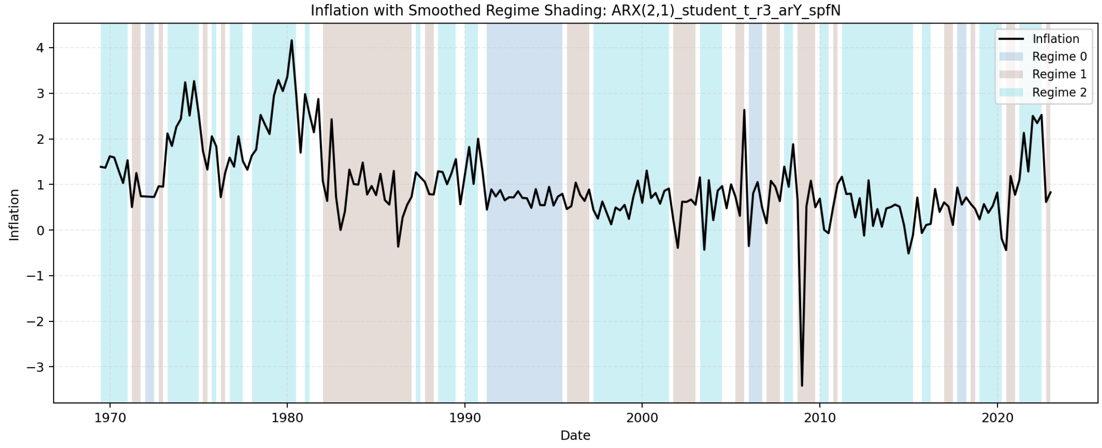
6.e. Transition Probabilities (Vector Form)

parameter	estimate
p[0->0]	0.832798
p[1->0]	0.0480447
p[2->0]	0.000150046
p[0->1]	0.165512
p[1->1]	0.646779
p[2->1]	0.24607

6.f. Transition Probability Matrix

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.832798	0.165512	0.00168973	1
Regime 1	0.0480447	0.646779	0.305177	1
Regime 2	0.000150046	0.24607	0.75378	1

6.g. Regime Classification Plot



7. BEGE GARCH Family

All BEGE specifications use the same effective sample, residual definition, mean-process menu, random-search protocol, density evaluator, and result selection rules. The model-specific shape recursions and restrictions are reported.

7.a. Mean Processes

The BEGE exercises estimate all four mean processes:

- **Constant**: $\pi_{t+1} = SPF_t + u_{t+1}$.
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + u_{t+1}$.
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + u_{t+1}$.
- **ARX(2,2)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_{t+1}$.

The mean-parameter bounds used in estimation and collection are:

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	—	—	—	—	—
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$[-1, 1]$	—	$[-10, 10]$	—
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$[-2, 2]$	$[-1, 1]$	$[-10, 10]$	—
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$[-2, 2]$	$[-1, 1]$	$[-10, 10]$	$[-10, 10]$

Best-model collection also imposes mean-process stationarity before a BEGE estimate is eligible for reporting. For ARX(1,1), this requires $| \rho_1 | < 1$. For ARX(2,1) and ARX(2,2), the stationarity condition is the usual AR(2) polynomial restriction:

$$1 - \rho_1 z - \rho_2 z^2 = 0 \tag{74}$$

must have both roots outside the unit circle. Equivalently, the reported estimate must satisfy

$$\rho_1 + \rho_2 < 1, \qquad \rho_2 - \rho_1 < 1, \qquad \rho_2 > -1. \tag{75}$$

Only the inflation-lag coefficients ρ_1 and ρ_2 enter this stationarity check; the SPF coefficients ϕ_1 and ϕ_2 are treated as exogenous-regressor loadings and are not included in the AR polynomial.

Starting values for the estimated mean parameters are drawn uniformly from OLS-centered intervals, $\hat{\theta}_j \pm 2\, SE\left(\hat{\theta}_j\right)$.

7.b. Residual Dynamics

Each BEGE model writes the inflation residual as

$$u_t = \sigma_p \omega_{p,t} - \sigma_n \omega_{n,t}, \qquad \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \quad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1), \tag{76}$$

where $\tilde{\Gamma}(a, 1)$ is a centered gamma shock with shape a and unit scale. Conditional on p_t and n_t,

- conditional variance: $\sigma_p^2 p_t + \sigma_n^2 n_t$,
- conditional skewness numerator: $2\left(\sigma_p^3 p_t - \sigma_n^3 n_t\right)$,
- conditional excess-kurtosis numerator: $6\left(\sigma_p^4 p_t + \sigma_n^4 n_t\right)$.

7.c. BEGE Volatility Specifications

The common parameter bounds used in estimation and result collection are:

Parameter group	Bound
Fixed shapes $\bar{p}, \bar{n}$ and intercepts p_0, n_0	$[0, 10]$
Persistence parameters ρ, ρ_p, ρ_n	$[0, 1]$
Shock-loading parameters $\phi^{\pm}, \phi_p, \phi_n, \phi_p^{\pm}, \phi_n^{\pm}$	$[0, 2]$
Scale parameters σ_p, σ_n	$[10^{-5}, 2]$

To rule out numerically degenerate and explosive variance paths, I use an EWMA path as a scale-adaptive reference for the BEGE implied variance. This is a loose numerical safeguard, not a variance-targeting restriction.[3] For residuals u_t, define

$$EWMA_1 = \frac{\sum_{j=1}^{\tau} \lambda^{j-1} u_j^2}{\sum_{j=1}^{\tau} \lambda^{j-1}}, \qquad EWMA_t = \lambda EWMA_{t-1} + (1-\lambda) u_{t-1}^2, \quad t \geq 2. \tag{77}$$

The EWMA implied-variance bounds are imposed during optimization and checked again during collection. With $\lambda = 0.94$ and $\tau = \min(75, T)$, let $s_u^2 := 1/T \sum_{j=1}^{T} u_j^2$ and $M_u := 1 + \max_{1 \leq s \leq T} u_s^2$. Then

$$\begin{aligned} \underline{v}_t &:= \max\{10^{-6} EWMA_t,\ 10^{-8} s_u^2\}, \\ \overline{v}_t &:= \max\{\min(10^{6} EWMA_t,\ 10^{7} M_u),\ M_u\}, \\ \underline{v}_t &\leq \sigma_p^2 p_t + \sigma_n^2 n_t \leq \overline{v}_t, \qquad t = 1, \dots, T. \end{aligned} \tag{78}$$

7.c.i. *Constant BEGE:*

The Constant BEGE model keeps both shape parameters fixed:

$$p_t = \bar{p}, \qquad n_t = \bar{n}. \tag{79}$$

Thus $u_t = \sigma_p \omega_{p,t} - \sigma_n \omega_{n,t}$ with $\omega_{p,t} \sim \tilde{\Gamma}(\bar{p}, 1)$ and $\omega_{n,t} \sim \tilde{\Gamma}(\bar{n}, 1)$.

7.c.ii. *Symmetric BEGE:*

The Symmetric BEGE model uses the same persistence and shock-loading parameters in both shape recursions:

$$\begin{aligned} p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{80}$$

The stability restriction is

$$\rho + \frac{\phi^+ + \phi^-}{2} < 1. \tag{81}$$

[3] The exponential weights follow the variance-forecasting idea in J.P. Morgan/Reuters, *RiskMetrics—Technical Document*, Fourth Edition (1996), Chapter 5, where recent squared shocks receive more weight than older shocks. Here the EWMA path is only a smoothed proxy for the local scale of u_t^2, not a BEGE variance target. The loose envelope is adapted from the `arch` package's `VolatilityProcess.variance_bounds`: the lower bound prevents the implied BEGE variance from collapsing toward zero, while the upper bound screens out explosive variance paths that would make likelihood evaluation numerically unreliable.

7.c.iii. *BadGood BEGE:*

The BadGood BEGE model lets each shape process react symmetrically to squared residuals, with separate good- and bad-environment parameters:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} u_{t-1}^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} u_{t-1}^2. \end{aligned} \tag{82}$$

The stability restrictions are

$$\rho_p + \phi_p < 1, \qquad \rho_n + \phi_n < 1. \tag{83}$$

7.c.iv. *Inflation/Deflation BEGE:*

The Inflation/Deflation BEGE model lets the good-news shape respond only to positive residuals and the bad-news shape respond only to negative residuals:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{84}$$

The stability restrictions are

$$\rho_p + \frac{\phi_p^+}{2} < 1, \qquad \rho_n + \frac{\phi_n^-}{2} < 1. \tag{85}$$

7.c.v. *Constant-p Full BEGE:*

Constant-p Full BEGE fixes the good-news shape and keeps the unrestricted Full BEGE update for the bad-news shape:

$$\begin{aligned} p_t &= p_0, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{86}$$

The dynamic shape process satisfies

$$\rho_n + \frac{\phi_n^+ + \phi_n^-}{2} < 1. \tag{87}$$

7.c.vi. *Constant-n Full BEGE:*

Constant-n Full BEGE fixes the bad-news shape and keeps the unrestricted Full BEGE update for the good-news shape:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0. \end{aligned} \tag{88}$$

The dynamic shape process satisfies

$$\rho_p + \frac{\phi_p^+ + \phi_p^-}{2} < 1. \tag{89}$$

7.c.vii. *Full BEGE:*

The Full BEGE model allows each shape process to respond separately to positive and negative residuals:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2. \end{aligned} \tag{90}$$

The stability restrictions are

$$\rho_p + \frac{\phi_p^+ + \phi_p^-}{2} < 1, \qquad \rho_n + \frac{\phi_n^+ + \phi_n^-}{2} < 1. \tag{91}$$

7.d. Random Search

BEGE estimation uses 50 independent seed jobs for each volatility specification. Within each seed job, each mean process uses 40 saved random draws, and each draw runs 25 optimizer starts. Therefore each BEGE specification and mean-process pair uses

$$50 \times 40 \times 25 = 50,000 \tag{92}$$

optimizer starts.

For volatility parameters, starting values are sampled inside the documented bounds. Shape intercept starts p_0, n_0 are sampled from the positive part of $[0, 10]$, scale starts σ_p, σ_n from $[10^{-5}, 2]$, and persistence and shock-loading starts are drawn so the relevant stability restriction is satisfied before optimization begins.

When a dynamic BEGE recursion is evaluated, I initialize the pre-sample p or n state by the parameter-implied unconditional backcast, $p_0/\left(1 - \rho - \frac{1}{2}(\phi^+ + \phi^-)\right)$ or the corresponding restricted formula, with a small positive numerical floor.[4] Constant-p and Constant-n models set the constant shape path directly.

7.e. Density Evaluation

Likelihood evaluation uses the stabilized BEGE density implementation documented in the BEGE Density section. The current evaluator keeps the closed-form hypergeometric expression in stable regions, switches to a guarded saddlepoint approximation when the closed form becomes numerically fragile.

7.f. Selection And Reporting

For each BEGE specification, the reported best-model page contains only the likelihood-best admissible estimate across mean processes. The by-mean best rows remain available in the CSV outputs linked from each report. Selection requires successful optimizer convergence, mean-process stationarity, documented parameter and stability constraints, and EWMA implied-variance bounds.

[4] The numerical floor is 10^{-4}: if the initial backcast or a later recursion produces a smaller value p_t, n_t, the code uses 0.0001 instead.

Each report also gives empirical 5%, median, and 95% quantiles for p_t, n_t, $\mathrm{Var}_t(u_t)$, the skewness numerator, and the excess-kurtosis numerator.

7.g. Standard Errors

I compute standard errors for the reported best estimate. For each observation, I approximate the score —the first derivative of that observation's log likelihood with respect to the parameters—using centered finite differences. I approximate the Hessian—the matrix of second derivatives of the total negative log likelihood—using the four-point centered finite-difference method implemented by `statsmodels.tools.numdiff.approx_hess`.

The default covariance is the sandwich matrix

$$\hat{V}\left(\hat{\theta}\right) = \hat{H}^{-1}\left(\sum_{t=1}^{T} \hat{g}_t \hat{g}_t{}'\right)\hat{H}^{-1}, \tag{93}$$

where $\hat{H}$ is the Hessian of the negative log likelihood and $\hat{g}_t$ is the numerical score contribution for observation t. If the sandwich calculation is numerically unstable, the collector uses an observed-information or inverse-OPG fallback. Standard errors that cannot be reliably identified are reported as `NA`.

8. BEGE Density

The BEGE density $f(u_t \mid p_t, n_t, \sigma_p, \sigma_n)$ is the conditional density of one inflation residual given the two gamma shape parameters and two scale parameters. This section compares three implementations:

- numerical integration, which is slow but useful as an independent benchmark;
- Justin's closed-form hypergeometric implementation;
- my stabilized modification on Justin's implementation.

The main conclusion is that Justin's closed-form formula is correct in stable regions, but its direct numerical evaluation can fail when the shape parameters enter the transition range where the BEGE distribution is already close to a Gaussian law. In those cases, very large terms in the hypergeometric expression nearly cancel. Finite-precision evaluation can then create artificial pointwise log densities that are numerically impossible and can dominate the likelihood search.

8.a. Executive Summary

- For moderate shape values, the current implementation and Justin's analytic implementation agree to numerical precision.
- On the fixed-shape timing tests below, the current implementation is about 1.5 times faster than Justin's implementation and about 957 times faster than 5,000-grid numerical integration at the median across the five test cases.
- The old job-side density produced implausibly large likelihoods on some dynamic BEGE estimates. For example, one BadGood ARX(2,1) row had stored LogLik 4920.7089, while the stabilized density gives −722.4342 and the saddlepoint check gives −722.7255.
- At the pointwise level, one old high-likelihood row reported $\ell_t = 90.3648$ for an observation with conditional variance 42.9373. The stabilized, saddlepoint, and Fourier-inversion values are all about −2.8008. This identifies a numerical failure of the hypergeometric evaluation, not an economic feature of the BEGE recursion.

8.b. What Changed In `BEGE_density.py`

Issue	Justin implementation	Current implementation	Reason
Hypergeometric evaluation	Direct `scipy.special.hyperu` or scalar `mpmath.hyperu` fallback.	Vectorized SciPy where stable, selective high-precision fallback, and log-domain asymptotic fallbacks when hyperu is nonfinite.	Avoid paying high-precision cost for every observation while keeping stable exact values.

Large-shape transition region	Uses the closed-form branch until a hard max-shape switch.	Guards exact values once $p_t + n_t \geq 40$, blends exact and saddlepoint between total shape 50 and 80, and replaces exact values when exact and saddlepoint differ by more than 2 log units.	The failure is driven by cancellation in total shape, and it can occur before a max-shape cutoff is reached.
Near-normal limit	No explicit analytic normal-limit rule.	Uses a Gaussian density when total shape is at least 500 and standardized skewness and excess kurtosis are both below 0.03.	Gamma shocks converge analytically to a normal law under variance-preserving shape rescaling.
Density continuity	Hard branch switches can create likelihood discontinuities.	Uses log-sum-exp blending in the transition band.	Keeps the objective smoother for optimization.
Failure diagnostics	High likelihoods can silently enter estimation results.	The collector recomputes stored estimates with the stabilized density and writes path/layer diagnostics to CSV.	Prevents stale or unstable density values from determining best models.

The current thresholds are deliberately conservative: exact hypergeometric values are still used when they agree with saddlepoint values, while suspicious large-shape exact values are replaced by the saddlepoint approximation.

I compare three fixed-shape BEGE density implementations on the ARX(1,1) residuals from the canonical effective sample. As a benchmark, the Gaussian OLS log-likelihood is −207.381. The residuals are mean-zero (0.00) with a standard deviation of 0.636326, a negative skewness of −1.047441 and a large excess kurtosis of 8.210174.

8.c. Shape Parameter Range

The range below uses eligible ARX(1,1) rows from the BadGood BEGE results only. For each saved parameter vector I recomputed the recursive shape paths on the common ARX(1,1) residuals, then summarized all observations and all eligible rows. The density comparison itself keeps the selected shape values constant across time.

Model	Rows	p q05	p median	p q95	n q05	n median	n q95	sigma_p med	sigma_n med
BadGood	994	1.4285	2.8130	6.4330	0.0714	0.3320	2.8342	0.2355	0.5555

8.d. *Fixed-Shape Parameter Sets*

The log likelihood and timing comparison uses the following five BadGood-based parameter sets. In each row, `sigma_p` and `sigma_n` are held at their BadGood medians.

Set	p	n	sigma_p	sigma_n
p q05, n median	1.428538	0.331970	0.235491	0.555532
p median, n median	2.813004	0.331970	0.235491	0.555532
p q95, n median	6.433015	0.331970	0.235491	0.555532
p median, n q05	2.813004	0.071355	0.235491	0.555532
p median, n q95	2.813004	2.834194	0.235491	0.555532

8.e. *Log Likelihood Comparison*

`BEGE_density.py` is the current implementation. `BEGE_density_Justin.py` is Justin's analytic formula, and the numerical columns use `BEGE_density_Numerical_Integration.py::loglikedgam_constant` at the stated grid size.

Parameter set	Current implementation	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	−214.162839	−214.162839	−223.900355	−213.330306	−214.116656	−214.148282	−214.146613	−214.160080
p median, n median	−188.957117	−188.957117	−196.801762	−188.438308	−189.359597	−189.033173	−188.947127	−188.954400
p q95, n median	−194.340481	−194.340481	−221.076698	−196.996227	−192.679189	−194.464527	−194.368904	−194.347495
p median, n q05	−212.877473	−212.877473	−380.654962	−214.556763	−214.448495	−212.976377	−213.323667	−212.904257
p median, n q95	−235.887717	−235.887717	−234.563988	−235.609633	−235.748009	−235.859932	−235.873999	−235.885265

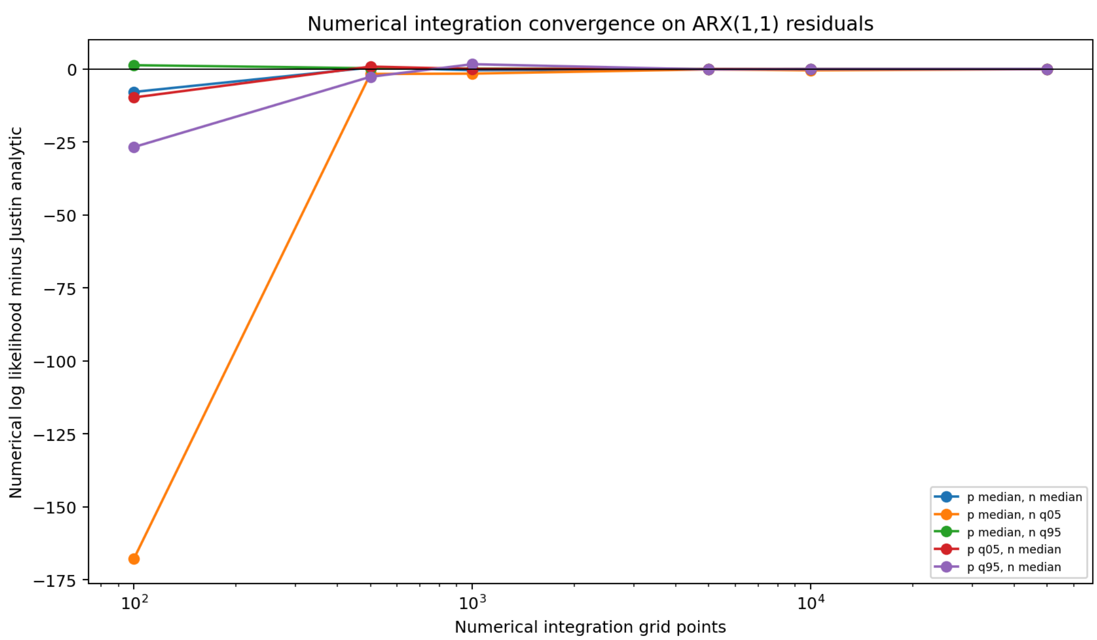
- The old numerical integration method converges toward Justin's likelihood function as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood.

8.f. *Evaluation Speed Comparison*

Parameter set	Current implementation	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	0.0003	0.0005	0.0086	0.0427	0.0883	0.4326	0.9102	4.5478

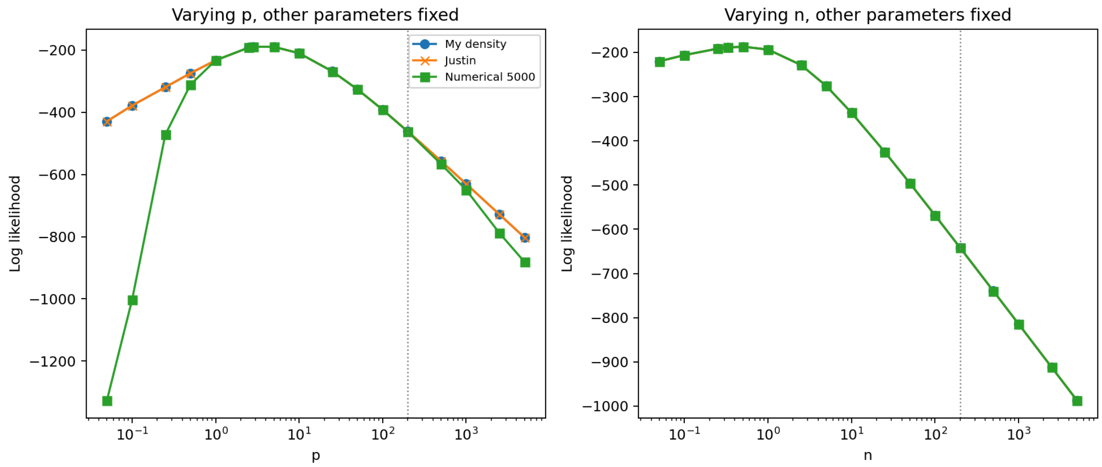
p median, n median	0.0004	0.0006	0.0075	0.0358	0.0730	0.3674	0.7613	3.8117
p q95, n median	0.0003	0.0005	0.0064	0.0308	0.0663	0.3258	0.6520	3.2415
p median, n q05	0.0003	0.0005	0.0076	0.0358	0.0737	0.3567	0.7479	3.6527
p median, n q95	0.0004	0.0006	0.0059	0.0284	0.0611	0.2900	0.5937	2.9724

- The current stabilized analytic implementation is slightly faster than Justin's direct analytic implementation in these moderate-shape cases (median speedup about 1.5 times).
- The current implementation is orders of magnitude faster than numerical integration. Relative to the 5,000-grid integration line, the median speedup is about 957 times across the five parameter sets.



8.g. Shape Tail Consistency

Holding the other parameters at the BadGood medians (`p=2.8130`, `n=0.3320`, `sigma_p=0.2355`, `sigma_n=0.5555`), I vary either `p` or `n` up to 5000 and compare the three density functions. The numerical integration line uses 5,000 grid points.



8.h. Robust Large-Shape Evaluation

The closed-form BEGE density contains a confluent hypergeometric term. It is accurate for small and moderate shape states, but it can become numerically fragile when the recursive shapes are large because several log terms nearly cancel. In that region, a low-precision evaluation can create artificial pointwise log densities that are orders of magnitude too high or too low.

For estimation and result collection, `BEGE_density.py` now uses a guarded large-shape rule:

- keep the exact hypergeometric expression for small-shape observations;
- use the cumulant saddlepoint density when total shape enters the fragile range, with a smooth transition rather than a single discontinuous cutoff;
- replace exact values by the saddlepoint value whenever the two disagree by more than the numerical tolerance in the guarded region;
- use the Gaussian density when total shape is large and standardized skewness and excess kurtosis are both numerically close to zero.

The saddlepoint approximation is based on the conditional cumulant-generating function of

$$u_t = \sigma_p(\Gamma(p_t, 1) - p_t) - \sigma_n(\Gamma(n_t, 1) - n_t). \tag{94}$$

It solves $K'(\hat{s}) = u_t$ and evaluates

$$\log f(u_t) \approx K(\hat{s}) - \hat{s}u_t - \frac{1}{2}\log\{2\pi K''(\hat{s})\}. \tag{95}$$

This construction respects the analytic normal limit of the BEGE distribution: as the gamma shapes grow while $\sigma_p^2 p_t + \sigma_n^2 n_t$ remains fixed, the standardized skewness and excess kurtosis shrink toward zero and the large-shape density converges to the Gaussian density with the same conditional variance.

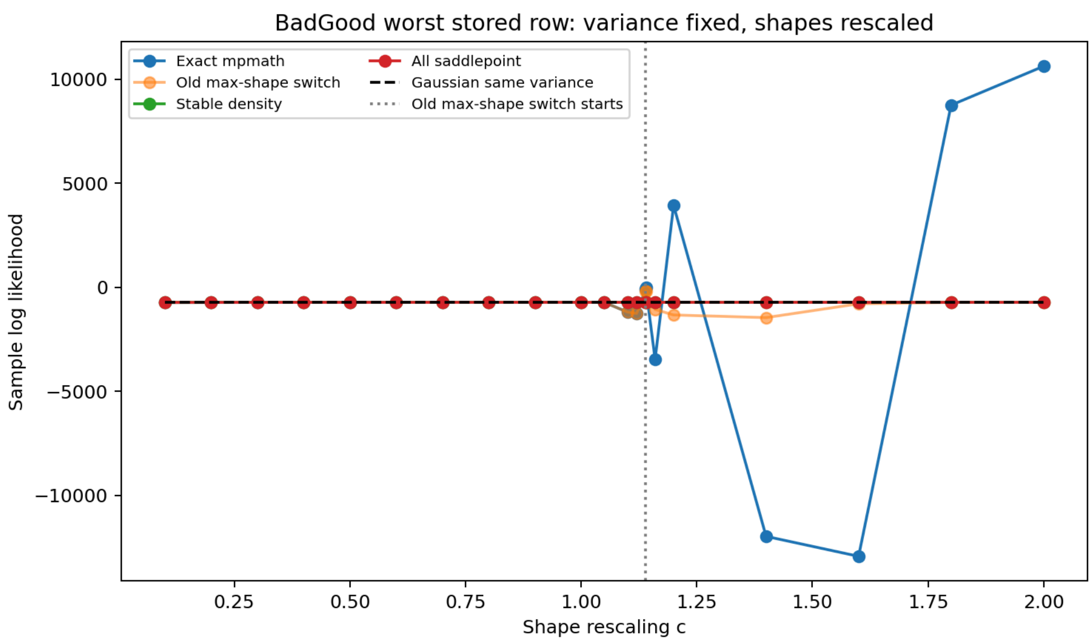
8.i. Saddlepoint Threshold Experiment

The BEGE distribution should approach a Gaussian law when both gamma shapes grow and the scale parameters shrink so that $\sigma_p^2 p_t + \sigma_n^2 n_t$ stays fixed. This check takes the same seed 45, draw 25, ARX(2,1) path and rescales

$$p_t(c) = cp_t, \qquad n_t(c) = cn_t, \qquad \sigma_p(c) = \sigma_p/\sqrt{c}, \qquad \sigma_n(c) = \sigma_n/\sqrt{c}. \tag{96}$$

The variance path is unchanged by construction. The exact hypergeometric branch is not monotonically better or worse as shapes grow: it is fragile in transition regions where large log terms cancel. A hard cutoff based only on `max(p_t,n_t)` is also too late. For this row, the old `max(p_t,n_t)=180` switch starts at $c = 1.1395$, while exact-branch failures already appear around $c = 1.10$ and the stored old job-side likelihood at $c = 1$ was 4920.7089.

Scale c	Median total shape	Max shape	Old max-shape saddle obs.	Exact mpmath LogLik	Old max-shape switch LogLik	Stabilized LogLik	Saddlepoint LogLik	Gaussian LogLik
0.1000	19.4893	15.7964	0	−728.2549	−728.2549	−728.2549	−727.1415	−720.5045
0.4000	77.9574	63.1856	0	−724.0184	−724.0184	−724.0388	−723.9614	−720.5045
0.8000	155.9147	126.3712	0	−722.6555	−722.6555	−722.6530	−722.9801	−720.5045
1.0000	194.8934	157.9640	0	−722.4380	−722.4380	−722.4342	−722.7255	−720.5045
1.1000	214.3827	173.7604	0	−1187.7835	−1187.7835	−722.3558	−722.6246	−720.5045
1.1395	222.0811	180.0000	1	−94.4486	−174.4248	−722.3280	−722.5884	−720.5045
1.2000	233.8721	189.5568	162	3934.4808	−1329.2867	−722.2885	−722.5364	−720.5045
1.8000	350.8081	284.3352	197	8749.6077	−722.0183	−722.0077	−722.1699	−720.5045
2.0000	389.7868	315.9280	200	10616.6236	−721.9500	−721.9408	−722.0858	−720.5045



The implemented rule therefore uses total shape, not a hard maximum-shape cap:

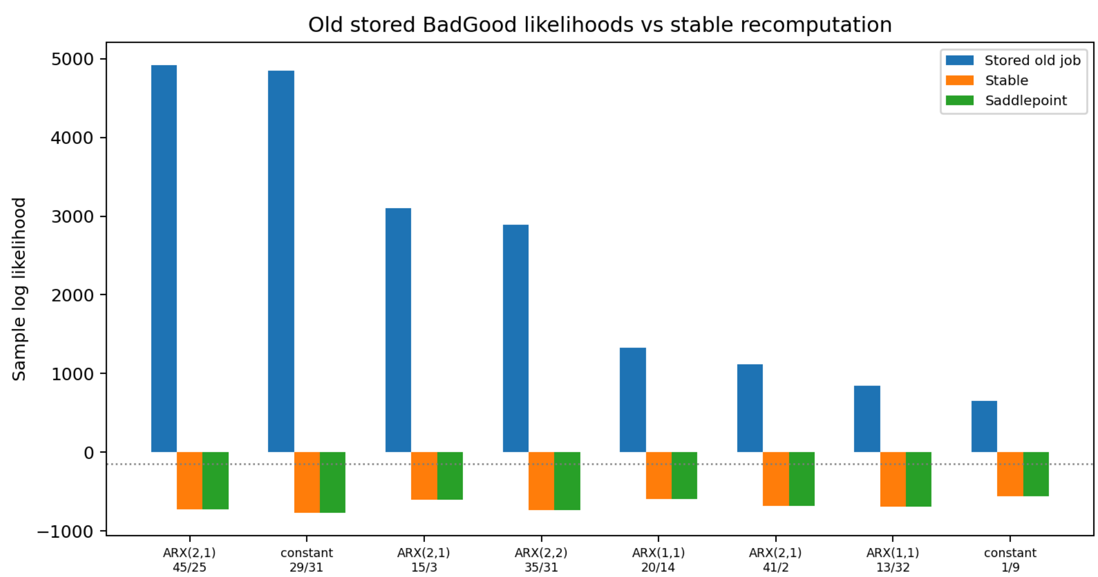
- guard exact hypergeometric values once `p_t+n_t >= 40`;
- smooth-blend exact and saddlepoint values between total shape 50 and 80;

- replace exact values when exact and saddlepoint differ by more than 2 log units in the guarded region;
- use the Gaussian limit only after total shape 500 and only when standardized skewness and excess kurtosis are both below 0.03 in absolute/magnitude terms.

8.j. Gigantic Stored Log-Likelihood Diagnostics

The table below starts from the BadGood results saved by the old production jobs. `Stored job LogLik` is the likelihood written by that old job-side density evaluation. The same parameter vectors are then recomputed on the same effective-sample residuals and recursive shape paths using the stabilized density, all-saddlepoint density, Gaussian density with the same conditional variance path, and reproducible exact branches available in the current git history.

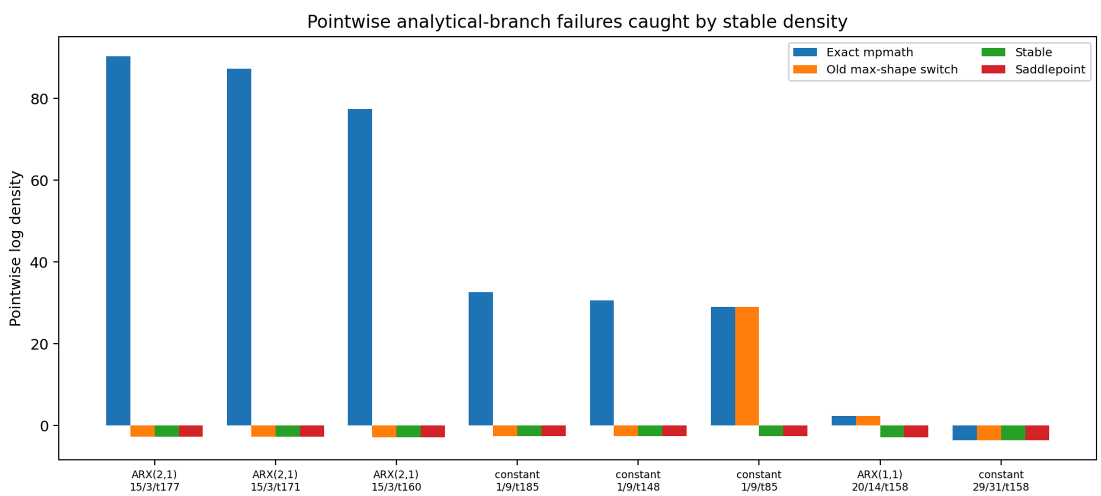
Mean	Seed	Draw	Stored job LogLik	Stabilized LogLik	Saddlepoint LogLik	Gaussian LogLik	Exact mpmath LogLik	Max shape	Median total shape	Median σ_t^2
ARX(2,1)	45	25	4920.7089	−722.4342	−722.7255	−720.5045	−722.4380	157.9640	194.8934	135.4737
constant	29	31	4846.4315	−770.7119	−770.6706	−770.2032	−1889.5667	171.9426	234.7388	225.1110
ARX(2,1)	15	3	3098.3394	−599.0093	−598.9692	−598.1718	−1186.8621	199.7122	220.0504	42.6849
ARX(2,2)	35	31	2892.0304	−734.4489	−734.2511	−730.1723	−807.8124	173.4795	205.7171	146.5852
ARX(1,1)	20	14	1330.8412	−590.1301	−590.0846	−592.7498	−584.8732	170.2031	206.8959	40.1486
ARX(2,1)	41	2	1113.7517	−684.6683	−684.6022	−680.7040	−684.8619	146.8413	164.9777	90.3592
ARX(1,1)	13	32	846.3034	−694.3325	−694.2827	−692.9410	−791.0434	156.0623	190.5299	104.1080
constant	1	9	651.3535	−561.1096	−561.0324	−561.6200	−1101.1363	209.9112	191.0321	30.1035



Across 8,000 BadGood rows in this diagnostic pass, the stored job-side likelihood produced 150 rows above `-150`, including 13 rows above zero. The stabilized recomputation produced zero rows above `-150`, and its largest value was −161.8929.

Some pointwise analytical-branch failures are reproducible from the current git history. The table reports the observations with the largest exact-vs-stable gaps among the old high-likelihood rows. The Fourier inversion column is a slow direct numerical check for the largest gaps.

Mean	Seed	Draw	Obs.	u_t	p_t	n_t	σ_t^2	Exact mpmath ℓ_t	Stabilized ℓ_t	Saddlepoint ℓ_t	Gaussian ℓ_t	Fourier ℓ_t
ARX(2,1)	15	3	177	0.0913	42.9229	187.7988	42.9373	90.3648	-2.8008	-2.8008	-2.7989	-2.8007
ARX(2,1)	15	3	171	0.2476	42.9393	189.9662	42.9916	87.2647	-2.8052	-2.8052	-2.8002	-2.8051
ARX(2,1)	15	3	160	0.1429	44.3305	199.7122	44.4494	77.4285	-2.8191	-2.8191	-2.8163	-2.8191
constant	1	9	185	-0.5080	188.8323	17.0463	30.2154	32.6685	-2.6480	-2.6480	-2.6274	-2.6503
constant	1	9	148	0.5209	180.3713	17.1093	30.2612	30.6176	-2.6069	-2.6069	-2.6284	-2.6090
constant	1	9	85	1.0855	174.4991	17.0414	30.1040	28.9611	-2.5957	-2.5957	-2.6408	-2.5979



These rows show why the huge stored likelihoods are not economically or numerically credible. A conditional variance around 30 to 44 cannot support pointwise log densities between 29 and 90. The stabilized values match saddlepoint and Fourier inversion, so the problem is numerical evaluation of the hypergeometric expression, not the BEGE recursion itself.

8.k. Findings

- BadGood quantiles provide the fixed-shape examples used in the likelihood and timing tables.
- The numerical integration function is useful as a convergence diagnostic, but it is much slower than the analytic functions and remains grid-sensitive.
- The shape-tail plot varies one shape parameter at a time while holding the other shape and both scale parameters at the BadGood medians.
- The saddlepoint switch should be based on guarded total shape and exact-vs-saddlepoint disagreement; `max(p_t,n_t)` is useful as a diagnostic but is not a reliable likelihood cutoff.
- Large-shape likelihood evaluation should not rely only on the closed-form hypergeometric expression. The guarded saddlepoint and Gaussian-limit checks prevent spurious gigantic likelihood values from determining BEGE estimates.

9. Constant BEGE Best Model Summary

Generated: `2026-06-18T10:10:04` Total estimations: `8000` Successful estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate across mean processes. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

9.a. *Selected Best Model*

Best admissible estimate ranked by log likelihood across mean processes.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	30	30	−181.6884	381.3768	411.7126

Empirical path quantiles:

Series	5%	Median	95%
p_t	2.6646	2.6646	2.6646
n_t	0.2821	0.2821	0.2821
σ_t^2	0.3952	0.3952	0.3952
s_t^2	−0.2286	−0.2286	−0.2286
k_t^2	0.9308	0.9308	0.9308

Mean process:

$$\pi_{t+1} = 0.0041 + 0.3232\,\pi_t + 0.1633\,\pi_{t-1} + 0.4107\,SPF_t + 0.1942\,SPF_{t-1} + u_{t+1} \tag{97}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.2712\,\omega_{p,t} - 0.8404\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(\bar{p}, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(\bar{n}, 1), \\ \bar{p} &= 2.6646, \qquad \bar{n} = 0.2821, \\ \mathrm{Var}_t(u_t) &= (0.2712)^2\,2.6646 + (0.8404)^2\,0.2821. \end{aligned} \tag{98}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.0041	0.1115
ρ_1	0.3232	0.1214

ρ_2	0.1633	0.0829
ϕ_1	0.4107	0.3014
ϕ_2	0.1942	0.2957
$\bar{p}$	2.6646	0.1839
$\bar{n}$	0.2821	0.1597
σ_p	0.2712	0.0218
σ_n	0.8404	0.3557

10. Symmetric BEGE Best Model Summary

Generated: `2026-06-18T10:10:15` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

10.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	35	12	−167.9012	359.8025	400.2501

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.6347	1.1647	5.6629
n_t	0.3243	0.5949	2.8917
σ_t^2	0.1638	0.3005	1.4610
s_t^2	−0.2097	−0.0432	−0.0235
k_t^2	0.1876	0.3441	1.6727

Mean process:

$$\pi_{t+1} = 0.2006 + 0.1978\,\pi_t + 0.2405\,\pi_{t-1} + 0.2257\,SPF_t + 0.1197\,SPF_{t-1} + u_{t+1} \tag{99}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.3592\,\omega_{p,t} - 0.5026\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \tag{100}$$

$$\begin{aligned} p_t &= 0.1762 + 0.6852\,p_{t-1} + \frac{0.4128}{2(0.3592)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0809}{2(0.3592)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.0900 + 0.6852\,n_{t-1} + \frac{0.4128}{2(0.5026)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0809}{2(0.5026)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{101}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.2006	0.0397
ρ_1	0.1978	0.0273
ρ_2	0.2405	0.0389
ϕ_1	0.2257	0.2154
ϕ_2	0.1197	0.1590
p_0	0.1762	0.0688
n_0	0.0900	0.0393
ρ	0.6852	0.0587
ϕ^+	0.4128	0.0560
ϕ^-	0.0809	0.0871
σ_p	0.3592	0.0670
σ_n	0.5026	0.0621

11. BadGood BEGE Best Model Summary

Generated: `2026-06-18T10:10:22` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

11.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,1)	39	14	−163.5594	351.1188	391.5664

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.4097	0.5799	1.7285
n_t	0.2985	0.7108	3.8717
σ_t^2	0.1416	0.2554	1.0320
s_t^2	−0.2196	0.0384	0.0683
k_t^2	0.1754	0.2991	1.1456

Mean process:

$$\pi_{t+1} = 0.2119 + 0.2240\,\pi_t + 0.2742\,\pi_{t-1} + 0.2699\,SPF_t + u_{t+1} \tag{102}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.4778\,\omega_{p,t} - 0.4042\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{103}$$

$$\begin{gathered} p_t = 0.1575 + 0.5795\,p_{t-1} + \frac{0.2143}{2(0.4778)^2}\,u_{t-1}^2, \\ n_t = 0.1900 + 0.2831\,n_{t-1} + \frac{0.6633}{2(0.4042)^2}\,u_{t-1}^2 \end{gathered} \tag{104}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.2119	0.0283
ρ_1	0.2240	0.0281
ρ_2	0.2742	0.0140
ϕ_1	0.2699	0.0447
p_0	0.1575	0.0329
n_0	0.1900	0.0469
ρ_p	0.5795	0.0474
ρ_n	0.2831	0.0315
ϕ_p	0.2143	0.0433
ϕ_n	0.6633	0.0058
σ_p	0.4778	0.0509
σ_n	0.4042	0.0328

12. Inflation/Deflation BEGE-GJR Best Model Summary

Generated: `2026-06-18T10:11:34` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

12.a. *Selected Best Model*

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	3	39	−167.9018	361.8035	405.6218

Empirical path quantiles:

Series	5%	Median	95%
p_t	4.5793	7.2032	28.6664
n_t	0.0584	0.0697	0.6744
σ_t^2	0.1699	0.2766	1.0806
s_t^2	−1.2028	−0.0927	0.0660
k_t^2	0.3575	0.4378	3.8913

Mean process:

$$\pi_{t+1} = 0.1246 + 0.2617\,\pi_t + 0.1743\,\pi_{t-1} + 0.2915\,SPF_t + 0.1749\,SPF_{t-1} + u_{t+1} \tag{105}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.1486\,\omega_{p,t} - 0.9893\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \tag{106}$$

$$\begin{aligned} p_t &= 1.2658 + 0.7076\,p_{t-1} + \frac{0.4250}{2(0.1486)^2}\left(u_{t-1}^{+}\right)^2, \\ n_t &= 0.0508 + 0.1317\,n_{t-1} + \frac{1.2626}{2(0.9893)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{107}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.1246	0.0726
ρ_1	0.2617	0.0767
ρ_2	0.1743	0.0758
ϕ_1	0.2915	0.3736
ϕ_2	0.1749	0.3151
p_0	1.2658	0.7514
n_0	0.0508	0.0486
ρ_p	0.7076	0.1141
ρ_n	0.1317	0.0583
ϕ_p^+	0.4250	0.1637
ϕ_n^-	1.2626	0.8901
σ_p	0.1486	0.0423
σ_n	0.9893	0.5903

13. Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:22` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

13.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	13	33	−165.2805	360.5611	411.1206

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.9806	1.1858	2.3678
n_t	0.0849	0.2556	2.8363
σ_t^2	0.1834	0.2969	1.1546
s_t^2	−0.9826	0.0427	0.1478
k_t^2	0.1925	0.3661	2.0843

Mean process:

$$\pi_{t+1} = 0.1937 + 0.1415\,\pi_t + 0.1380\,\pi_{t-1} + 0.2283\,SPF_t + 0.3635\,SPF_{t-1} + u_{t+1} \tag{108}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.3802\,\omega_{p,t} - 0.5787\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \tag{109}$$

$$\begin{aligned} p_t &= 0.0347 + 0.9561\,p_{t-1} + \frac{0.0000}{2(0.3802)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0334}{2(0.3802)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.0295 + 0.5611\,n_{t-1} + \frac{0.7784}{2(0.5787)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0720}{2(0.5787)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{110}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.1937	0.0783
ρ_1	0.1415	0.0611
ρ_2	0.1380	0.0618
ϕ_1	0.2283	0.2319
ϕ_2	0.3635	0.1626
p_0	0.0347	0.0277
n_0	0.0295	0.0258
ρ_p	0.9561	0.0242
ρ_n	0.5611	0.0879
ϕ_p^+	0.0000	NA
ϕ_p^-	0.0334	0.0159
ϕ_n^+	0.7784	0.1702
ϕ_n^-	0.0720	0.0947
σ_p	0.3802	0.0621
σ_n	0.5787	0.1108

14. Constant-p Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:11` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

14.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	28	25	−166.7909	357.5819	398.0296

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.4727	0.4727	0.4727
n_t	0.4590	0.9268	7.3165
σ_t^2	0.1990	0.2674	1.2023
s_t^2	−0.6797	0.0355	0.0878
k_t^2	0.2795	0.3396	1.1602

Mean process:

$$\pi_{t+1} = 0.1831 + 0.1660\,\pi_t + 0.0787\,\pi_{t-1} + 0.2999\,SPF_t + 0.3076\,SPF_{t-1} + u_{t+1} \tag{111}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.5281\,\omega_{p,t} - 0.3825\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \tag{112}$$

$$\begin{aligned} p_t &= 0.4727, \\ n_t &= 0.2113 + 0.4747\,n_{t-1} + \frac{0.7577}{2(0.3825)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.2362}{2(0.3825)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{113}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.1831	0.0290

ρ_1	0.1660	0.0376
ρ_2	0.0787	0.0273
ϕ_1	0.2999	0.1631
ϕ_2	0.3076	0.1028
p_0	0.4727	0.1424
n_0	0.2113	0.0517
ρ_n	0.4747	0.0291
ϕ_n^+	0.7577	0.0098
ϕ_n^-	0.2362	0.0348
σ_p	0.5281	0.1365
σ_n	0.3825	0.0364

15. Constant-n Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:30` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

15.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	31	40	−170.2704	364.5408	404.9885

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.6223	1.1422	5.6821
n_t	0.4055	0.4055	0.4055
σ_t^2	0.2157	0.2840	0.8798
s_t^2	−0.0950	−0.0455	0.3862
k_t^2	0.3302	0.3840	0.8532

Mean process:

$$\pi_{t+1} = 0.1460 + 0.1849\,\pi_t + 0.0812\,\pi_{t-1} + 0.9787\,SPF_t + -0.3699\,SPF_{t-1} + u_{t+1} \tag{114}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3623\,\omega_{p,t} - 0.5750\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{115}$$

$$\begin{aligned} p_t &= 0.1443 + 0.7377\,p_{t-1} + \frac{0.4074}{2(0.3623)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0184}{2(0.3623)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.4055 \end{aligned} \tag{116}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.1460	0.0336

ρ_1	0.1849	0.0432
ρ_2	0.0812	0.0439
ϕ_1	0.9787	0.1048
ϕ_2	−0.3699	0.0891
p_0	0.1443	0.0321
n_0	0.4055	0.0525
ρ_p	0.7377	0.0172
ϕ_p^+	0.4074	0.0285
ϕ_p^-	0.0184	0.0323
σ_p	0.3623	0.0408
σ_n	0.5750	0.0736